

❑ **Time remaining in the project:** 8 months

❑ **Completed parts:**

Compared with the optimization part in the document, the following contents have been included in our model

- Start from existing network
- Objective
 - Minimize cost (calculation explained in the sections below)
 - Chartering
 - Bunkering
 - Port call cost
 - Transshipments
- Constraints
 - Base constraints
 - Satisfy all demand (soft)
 - Vessel capacities
 - Respect cargo transit time (soft)
 - Cargo value fully depreciates if the route it takes is tardy (by a fraction of its expected transit time)
 - Max. number of operations performed at each port is upper-bounded by (port stay time/productivity)
 - Vessel speed must be between 10kts (hard) and 16.5kts (soft)
 - Steaming above 16.5kts possible, but heavily penalized
 - Additional constraints for lines
 - Number of times a port can be visited in the same Service ≤ 3
 - No repeating of sub-routes within a service's port rotation
 - E.g. CNXMN – TWKHH – TWTXG – CNXMN – TWKHH not allowed
 - Number of unique ports in a service ≤ 20 ,
 - We can force longer or shorter services to explore new ideas
- Actions Possible
 - Addition of a regular/transshipment port call on a service
 - Option 1: $A \rightarrow B \rightarrow C$ becomes $A \rightarrow B \rightarrow X \rightarrow C$
 - Option 2: $A \rightarrow B \rightarrow C$ becomes $A \rightarrow B \rightarrow X \rightarrow B \rightarrow C$ → Can be achieved by updating multiple actions in our model
 - Removal/Change of a regular/transshipment port on a service →
 - Shifting a port call from one service to another (can also be done on a single service, i.e. “sliding” the position of the port call) →
 - Swapping the exact positions of 2 port calls (can be done between distinct lines, or on the same line) →

Small-scale trials (optimizing the shipping network in Southeast Asia) have been completed.

❑ Uncompleted parts:

- Constraints
 - Additional constraints for lines
 - Line will be penalized if the buffer is above 30%
 - Number of ports in a Service that are called more than once ≤ 2
 - Minimum 3h berthing time
 - A route cannot come in and out of APAC
 - Max direct port to port connection = 2k miles
 - Every line must visit at least one TS port
 - All POL-POD pairs (in the demand) must be connected by no more than 3 transshipments
 - Frozen lines
 - These lines are not fully operated by CNC and cannot be modified.
 - Port rotation and speed can't be modified
 - Flow on them can be modified.
 - Additional constraints for flow
 - Cargo can be transshipped only at select ports (including ports that carried out intra-Asia transshipments in the demand routing data)
 - There needs to be at least 1 day difference between ETB2 and ETD1 for a transshipment between 2 lines to be possible
 - Singapore transshipments must be at least 72.5% of SG/MY/ID transshipments
- Actions Possible
 - Change vessel size rank
 - Change vessel speed between port
 - Change the round-trip duration (and number of vessels deployed) of the service

} These actions will be implemented in the **scheduling model** in Tier 2

❑ Timeline:

	2025.9	2025.10	2025.11	2025.12	2026.1	2026.2	2026.3	2026.4
Add additional constraints and update the model								
Complete large-scale trials								
Finish the scheduling model in Tier 2								